Mathematics Higher

Instructions to Markers

1. Marks must be assigned in accordance with these marking instructions. In principle, marks are awarded for what is correct, rather than marks deducted for what is wrong.

2. Award one mark for each 'bullet' point. Each error should be underlined in RED at the point in the working where it first occurs, and not at any subsequent stage of the working.

3. The working subsequent to an error must be followed through by the marker with possible full marks for the subsequent working, provided that the difficulty involved is approximately similar. Where, subsequent to an error, the working is eased, a deduction(s) of marks(s) should be made.

   This may happen where a question is divided into parts. In fact, failure to even answer an earlier section does not preclude a candidate from assuming the result of that section and obtaining full marks for a later section.

4. Correct working should be ticked (✓). This is essential for later stages of the SQA procedures. Where working subsequent to an error(s) is correct and scores marks, it should be marked with a crossed tick (✓ or ✗ ✓). In appropriate cases attention may be directed to work which is not quite correct (eg bad form) but which has not been penalised, by underlining with a dotted or wavy line.

   Work which is correct but inadequate to score any marks should be corrected with a double cross tick (✓).

5. - The total mark for each section of a question should be entered in red in the **outer** right hand margin, opposite the end of the working concerned.
   - Only the mark should be written, **not** a fraction of the possible marks.
   - These marks should correspond to those on the question paper and these instructions.

6. It is of great importance that the utmost care should be exercised in adding up the marks. Where appropriate, all summations for totals and grand totals must be carefully checked.

   Where a candidate has scored zero marks for any question attempted, "0" should be shown against the answer.

7. As indicated on the front of the question paper, full credit should only be given where the solution contains appropriate working. Accept answers arrived at by inspection or mentally where it is possible for the answer so to have been obtained. Situations where you may accept such working will normally be indicated in the marking instructions.

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8. Do not penalise:

- working subsequent to a correct answer
- legitimate variations in numerical answers
- correct working in the "wrong" part of the question
- omission of units
- bad form

9. No piece of work should be scored through without careful checking – even where a fundamental misunderstanding is apparent early in the answer. Reference should always be made to the marking scheme – answers which are widely off-beam are unlikely to include anything of relevance but in the vast majority of cases candidates still have the opportunity of gaining the odd mark or two provided it satisfies the criteria for the mark(s).

10. If in doubt between two marks, give an intermediate mark, but without fractions. When in doubt between consecutive numbers, give the higher mark.

11. In cases of difficulty covered neither in detail nor in principle in the Instructions, attention may be directed to the assessment of particular answers by making a referral to the PA. Please see the general instructions for PA referrals.

12. No marks should be deducted at this stange for careless or badly arranged work. In cases where the writing or arrangement is very bad, a note may be made on the upper left-hand corner of the front cover of the script.

13. Transcription errors: In general, as a consequence of a transcription error, candidates lose the opportunity of gaining either the first ic mark or the first pd mark.

14. Casual errors: In general, as a consequence of a casual error, candidates lose the opportunity of gaining either the first ic mark or the first pd mark.

15. **Do not write any comments on the scripts.** A **revised** summary of acceptable notation is given on page 4.

Summary

Throughout the examination procedures many scripts are remarked. It is essential that markers follow common procedures:

1. **Tick** correct working.
2. Put a mark in the **right-hand margin to match the marks allocations on the question paper.**
3. Do **not** write marks as fractions.
4. Put each mark **at the end** of the candidate's response to the question.
5. **Follow through** errors to see if candidates can score marks subsequent to the error.
6. Do **not** write any comments on the scripts.

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Higher Mathematics: A Guide to Standard Signs and Abbreviations

Remember – No comments on the scripts. Please use the following and nothing else.

Signs

✓	The tick. You are not expected to tick every line but of course you must check through the whole of a response.	Bullets showing where marks have been allotted may be shown on scripts		margins
____ ✗	The cross and underline. Underline an error and place a cross at the end of the line.	$\frac{dy}{dx} = 4x - 7$ ✓• $4x - 7 = 0$ ✗ $x = \frac{7}{4}$ $y = 3\frac{7}{8}$ ✓•		2
✓ or ✗ ✓	The tick-cross. Use this to show correct work where you are **following through** subsequent to an error.	$C = (1,-1)$ ✗ $m = \frac{3-(-1)}{4-1}$ $m_{rad} = \frac{4}{3}$ ✓• $m_{tgt} = \frac{-1}{\frac{4}{3}}$ $m_{tgt} = -\frac{3}{4}$ ✓• $y - 3 = -\frac{3}{4}(x-2)$ ✓•		3
^	The roof. Use this to show something is missing such as a crucial step in a proof of a 'condition' etc.	$x^2 - 3x = 28$ ✓• $x = 7$ ✓✓		1
~	The tilde. Use this to indicate a minor transgression which is not being penalised (such as bad form).	$\sin(x) = 0 \cdot 75 = inv\sin(0 \cdot 75) = 48 \cdot 6^\circ$ ✓•		1
✓✓	The double-cross tick. Use this to show correct work but which is inadequate to score any marks. This may happen when working has been eased.			

Remember – No comments on the scripts. No abbreviations. No new signs.
Please use the above and nothing else.

All of these are to help us be more consistent and accurate.

Note: There is no such thing as a transcription error, a trivial error, a casual error or an insignificant error. These are all mistakes and as a consequence a mark is lost.

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1 Find $\int \frac{4x^3 - 1}{x^2}\,dx, \quad x \neq 0$. **4**

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
1		4	C	C14, C13	CN	05/20

The primary method m/s is based on the following generic m/s. THIS GENERIC M/S MAY BE USED AS AN EQUIVALENCE GUIDE BUT ONLY WHERE A CANDIDATE DOES NOT USE THE PRIMARY METHOD OR ANY ALTERNATIVE METHOD SHOWN IN DETAIL IN THE MARKING SCHEME

- $\bullet^1$ ss: arrange in integrable form
- $\bullet^2$ pd: integrate positive index
- $\bullet^3$ pd: integrate negative index
- $\bullet^4$ ic: complete including const. of int.

Primary Method : Give 1 mark for each •

- $\bullet^1$ $4x - x^{-2}$
- $\bullet^2$ $\frac{4x^2}{2}$
- $\bullet^3$ $-\frac{x^{-1}}{-1}$
- $\bullet^4$ $2x^2 + x^{-1} + c$ *[Note 3]*

4 marks

Notes

1 If incorrectly expressed in integrable form, follow throughs must match the generic marking scheme.
2 **•3** can only be awarded on follow through provided the integral involves a negative index.
3 **•4** can only be awarded if the constant of integration appears somewhere in the working.
4 **•4** can only be awarded as a result of at least one valid integration at the **•2** or **•3** stage.

Common Error 1

- $\bullet^1$ $\times$ $4x - 1$
- $\bullet^2$ $\times\surd$ $2x^2$
- $\bullet^3$ $\times$ $-x$ $[see\ Generic\ \bullet^3]$
- $\bullet^4$ $\times\surd$ $2x^2 - x + c$

award 2 marks

Common Error 2

- $\bullet^1$ $\times$ $4x^3 - 1 - x^{-2}$
- $\bullet^2$ $\times\surd$ $\frac{4x^4}{4} - x$
- $\bullet^3$ $\times\surd$ $-\frac{x^{-1}}{-1}$
- $\bullet^4$ $\times\surd$ $x^4 - x + x^{-1} + c$

award 3 marks

Common Error 3

- $\bullet^1$ $\times$ $4x^3 - 1 + x^{-2}$
- $\bullet^2$ $\times\surd$ $\frac{4x^4}{4} - x$
- $\bullet^3$ $\times\surd$ $\frac{x^{-1}}{-1}$
- $\bullet^4$ $\times\surd$ $x^4 - x - x^{-1} + c$

award 3 marks

Common Error 4

$\frac{x^4 - x}{\frac{1}{2}x^2} + c$

award 0 marks

Common Error 5

$(4x - 1)x^{-2}$
$(2x^2 - x)(-x^{-1}) + c$

award 0 marks

Common Error 6

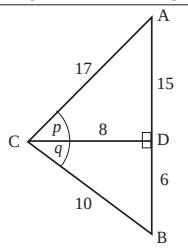
$(4x^3 - 1)x^{-2}$
$(x^4 - x)(-x^{-1}) + c$

award 0 marks

Throughout this paper, unless specifically mentioned, a correct answer with no working receives no credit.

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2 Triangles ACD and BCD are right-angled at D with angles p and q and lengths as shown in the diagram.



(a) Show that the exact value of $\sin(p+q)$ is $\frac{84}{85}$. 4

(b) Calculate the exact values of

(i) $\cos(p+q)$ (ii) $\tan(p+q)$. 3

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
2	a	4	C	T9	CN	05/41
	b	3	C	T9	CN	

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- •1 ic: interpret diagram
- •2 ic: interpret diagram
- •3 ss: expand sin(A+B)
- •4 pd: sub. and complete
- •5 ss: expand cos(A+B)
- •6 pd: sub. and complete
- •7 ic: use $\tan(x) = \sin(x) / \cos(x)$

Primary Method : Give 1 mark for each •

- •1 $\cos(p) = \frac{8}{17}, \sin(p) = \frac{15}{17}$ ***[Note 1]***
- •2 $\cos(q) = \frac{8}{10}, \sin(q) = \frac{6}{10}$ **stated or implied by •4 when written in the same order as •3**
- •3 $\sin(p)\cos(q) + \cos(p)\sin(q)$ **explicitly stated**
- •4 $\frac{15}{17} \times \frac{8}{10} + \frac{8}{17} \times \frac{6}{10} =$ & *complete* **4 marks**

- •5 $\cos(p)\cos(q) - \sin(p)\sin(q)$
- •6 $-\frac{13}{85}$ *or equivalent fraction*
- •7 $-\frac{84}{13}$ *or equivalent fraction* $\left(eg\ -\frac{7140}{1105}\right)$ **3 marks**

Notes

1 **•1** and **•2** may, if necessary, be awarded as follows

- •1 $\sin(p) = \frac{15}{17}, \sin(q) = \frac{6}{10}$
- •2 $\cos(p) = \frac{8}{17}, \cos(q) = \frac{8}{10}$

2 For **•4**
There has to be some working to show the completion.

eg $\ldots\ldots\ldots = \frac{120+48}{170} = \frac{168}{170} = \frac{84}{85}$

or

$\ldots\ldots\ldots = \frac{60}{85} + \frac{24}{85} = \frac{84}{85}$

or

$\ldots\ldots\ldots = \frac{12}{17} + \frac{24}{85} = \frac{84}{85}$

3 Calculating approx angles using invsin and invcos can gain no credit at any point.

4 Any attempt to use $\sin(p+q) = \sin(p) + \sin(q)$ **loses •3 and •4.**
Any attempt to use $\cos(p+q) = \cos(p) + \cos(q)$ **loses •5 and •6.**
This second option must not be treated as a repeated error.

Alternative 1 (for marks 3 & 4)

- •3 $\frac{21}{\sin(p+q)} = \frac{10}{\frac{8}{17}}$
- •4 $10\sin(p+q) = \frac{168}{17}$ *and complete*

Alternative 2 (for marks 5 & 6)

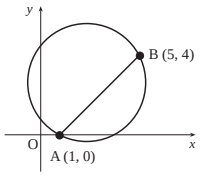
- •5 $\cos(p+q) = \frac{17^2 + 10^2 - 21^2}{2.17.10}$
- •6 $-\frac{13}{85}$

Alternative 3 (for marks 5 & 6)

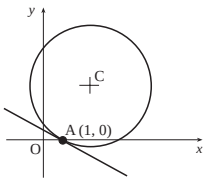
- •5 $\cos^2(p+q) = 1 - \left(\frac{84}{85}\right)^2$
- •6 $\cos(p+q) = -\frac{13}{85}$ with justification of the choice of negative sign
  e.g. $(15+6)^2\ (= 441) > 17^2 + 10^2 (= 389)$
  or using the cosine rule

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3 (a) A chord joins the points A(1, 0) and B(5, 4) on the circle as shown in the diagram.
Show that the equation of the perpendicular bisector of chord AB is $x + y = 5$. **4**



(b) The point C is the centre of this circle. The tangent at the point A on the circle has equation $x + 3y = 1$.
Find the equation of the radius CA. **4**



(c) (i) Determine the coordinates of the point C.
(ii) Find the equation of the circle. **4**

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
3	a	4	C	G7	CN	05/44
	b	4	C	G15	CN	
	c	4	C	G10	CN	

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- •1 ss: find perp. bisector
- •2 pd: calc. perp. gradient
- •3 ss: find approp. mid-point
- •4 ic: complete proof
- •5 ss: compare with $y = mx + c$
- •6 ic: state gradient
- •7 ss: find gradient of radius
- •8 ic: state equation of line
- •9 ss: solve sim. equations
- •10 pd: solve sim. equations
- •11 ic: state equation of circle
- •12 pd: calculate radius

Primary Method : Give 1 mark for each •

- •1 $m_{AB} = 1$
- •2 $m_{\perp} = -1$
- •3 midpoint $= (3, 2)$
- •4 $y - 2 = -1(x - 3)$ *and* complete *[Notes 1,2]* **4 marks**
- •5 $y = -\frac{1}{3}x \ldots\ldots$ **stated/implied by •6**
- •6 $m_{tgt} = -\frac{1}{3}$
- •7 $m_{rad} = 3$ **stated/implied by •8**
- •8 $y - 0 = 3(x - 1)$ *[Note 3]* **4 marks**
- •9 *use* $x + y = 5$ *and* $y = 3x - 3$ *[Notes 4,5]*
- •10 $x = 2,\ y = 3$
- •11 $(x - 2)^2 + (y - 3)^2 = r^2$
- •12 $r^2 = 10$ *[Note 6]* **4 marks**

Notes

1 To gain **•4** some evidence of completion needs to be shown

eg $y - 2 = -1(x - 3)$
$y - 2 = -x + 3$
$y + x = 5$

2 **•4** is only available if an attempt has been made to find and use both a perpendicular gradient and a midpoint.

3 **•8** is only available if an attempt has been made to find and use a perpendicular gradient.

4 At the **•9, •10** stage
Guessing (2,3) (from stepping) and checking it lies on perp. bisector of AB may be awarded •9 and •10
Guessing (2,3) (with or without reason) and with no check gains **neither** •9 nor •10

5 Solving $y = 3x - 3$ *and* $x + 3y = 1$ leading to (1,0) will lose **•9 and •10**.

6 to gain **•12** some evidence of use of the distance formula needs to be shown.

7 At the **•11 and •12** stage
Subsequent to a guess for the coordinates of C, **•11 and •12** are only available if the guess is such that 0<x<5 and 0<y<4.

Alternative 1 [for •9 and •10]

- •9 D=(3,6) where D is intersection of the perp. to AB through B and the circle.
- •10 C = midpoint of AD = (2,3)

Common Error 1 [for •5 to •8]

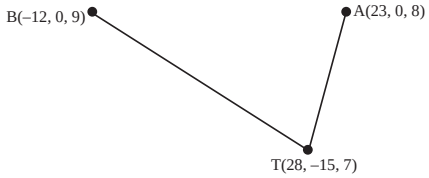
$3y = -x + 1$
$m = -1$
$m_{rad} = 1$
$y - 0 = 1(x - 1)$
•5 × •6 × •7 × *eased* •8 ×√
award 1 mark

Common Error 2 [for •5 to •8]

$x + 3y = 1$ *so* $m = 3$
$y - 0 = 3(x - 1)$
award 0 marks

7

4 The sketch shows the positions of Andrew(A), Bob (B) and Tracy(T) on three hill-tops.
Relative to a suitable origin, the coordinates (in hundreds of metres) of the three people are A(23, 0, 8), B(–12, 0, 9) and T(28, –15, 7).
In the dark, Andrew and Bob locate Tracy using heat-seeking beams.



(*a*) Express the vectors $\overrightarrow{TA}$ and $\overrightarrow{TB}$ in component form. **2**

(*b*) Calculate the angle between these two beams. **5**

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
4	a	2	C	G17	CN	05/55
	b	5	C	G28	Ca	

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- $\bullet^1$ ic: state vector components
- $\bullet^2$ ic: state vector components
- $\bullet^3$ pd: find length of vector
- $\bullet^4$ pd: find length of vector
- $\bullet^5$ pd: find scalar product
- $\bullet^6$ ss: use scalar product
- $\bullet^7$ pd: evaluate angle

Primary Method : Give 1 mark for each •

- $\bullet^1$ $\vec{TA} = \begin{pmatrix} -5 \\ 15 \\ 1 \end{pmatrix}$
- $\bullet^2$ $\vec{TB} = \begin{pmatrix} -40 \\ 15 \\ 2 \end{pmatrix}$ *[Notes 1,2]* **2 marks**
- $\bullet^3$ $|\vec{TA}| = \sqrt{251}$
- $\bullet^4$ $|\vec{TB}| = \sqrt{1829}$
- $\bullet^5$ $\vec{TA}.\vec{TB} = 427$
- $\bullet^6$ $\cos(A\hat{T}B) = \dfrac{\vec{TA}.\vec{TB}}{|\vec{TA}||\vec{TB}|}$ **stated or implied by •7** *[Note 3]* **5 marks**
- $\bullet^7$ $A\hat{T}B = 50 \cdot 9° \ \ OR \ \ 0.889^c \ \ OR \ 56.6$ grads *[Note 4]*

Notes
In (a)

1 For calculating $\overrightarrow{AT}$ *and* $\overrightarrow{BT}$ award 1 mark out of 2.
2 Treat column vectors written like (–40, 15, 2) as bad form.

In (b)

3 For candidates who do not attempt **•7**, the formula quoted at **•6** must relate to the labelling in the question for **•6** to be awarded.
4 Do not penalise premature rounding.
5 The use of $\tan(A\hat{T}B) = \dfrac{\vec{TA}.\vec{TB}}{|\vec{TA}||\vec{TB}|}$ loses **•6**
6 The use of $\cos(A\hat{T}B) = \dfrac{\vec{TA}.\vec{TB}}{|\vec{AB}|}$ means that only **•5** and **•7** are available.

Alternative 1 for •3 to •7 (Cosine Rule)

- $\bullet^3$ $\left|\vec{TA}\right| = \sqrt{251}$
- $\bullet^4$ $\left|\vec{TB}\right| = \sqrt{1829}$
- $\bullet^5$ $\left|\vec{AB}\right| = \sqrt{1226}$
- $\bullet^6$ $\cos(A\hat{T}B) = \dfrac{1829 + 251 - 1226}{2.\sqrt{1829}.\sqrt{251}}$ **stated or implied by •7**
- $\bullet^7$ $A\hat{T}B = 50 \cdot 9°$

Common Error No.1

- $\bullet^1$ × $\overrightarrow{TA} = t - a = \begin{pmatrix} 5 \\ -15 \\ -1 \end{pmatrix}$
- $\bullet^2$ ×√ $\vec{TB} = t - b = \begin{pmatrix} 40 \\ -15 \\ -2 \end{pmatrix}$

award 1 mark

Common Error No.2

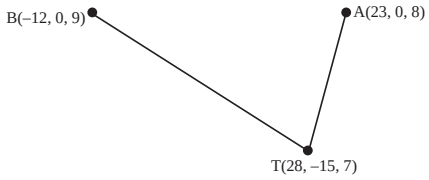
- $\bullet^1$ × $\overrightarrow{TA} = t + a = \begin{pmatrix} 51 \\ -15 \\ 15 \end{pmatrix}$
- $\bullet^2$ ×√ $\vec{TB} = t + b = \begin{pmatrix} 16 \\ -15 \\ 16 \end{pmatrix}$

award 1 mark

Further common errors overleaf.

8

4 The sketch shows the positions of Andrew(A), Bob (B) and Tracy(T) on three hill-tops.
Relative to a suitable origin, the coordinates (in hundreds of metres) of the three people are A(23, 0, 8), B(–12, 0, 9) and T(28, –15, 7).
In the dark, Andrew and Bob locate Tracy using heat-seeking beams.



(*a*) Express the vectors $\overrightarrow{TA}$ and $\overrightarrow{TB}$ in component form. **2**

(*b*) Calculate the angle between these two beams. **5**

Common Error 1 : Finding angle BOA

using $\vec{OB} = \begin{pmatrix} -12 \\ 0 \\ 9 \end{pmatrix}$ *and* $\vec{OA} = \begin{pmatrix} 23 \\ 0 \\ 8 \end{pmatrix}$

- $|\vec{OB}| = \sqrt{225}$ *and* $|\vec{OA}| = \sqrt{593}$
- $\vec{OB}.\vec{OA} = -204$
- $\cos(B\hat{O}A) = \dfrac{\vec{OB}.\vec{OA}}{|\vec{OB}||\vec{OA}|}$
- $B\hat{O}A = 124{\cdot}0^\circ$ *OR* $2{\cdot}163^c$

award 1 mark per bullet

Common Error 2 : Finding angle BOT

using $\vec{OB} = \begin{pmatrix} -12 \\ 0 \\ 9 \end{pmatrix}$ *and* $\vec{OT} = \begin{pmatrix} 28 \\ -15 \\ 7 \end{pmatrix}$

- $|\vec{OB}| = \sqrt{225}$ *and* $|\vec{OT}| = \sqrt{1058}$
- $\vec{OB}.\vec{OT} = -273$
- $\left\langle \begin{array}{l} \cos(B\hat{O}T) = \dfrac{\vec{OB}.\vec{OT}}{|\vec{OB}||\vec{OT}|} \\ B\hat{O}T = 124{\cdot}0^\circ \ \ OR \ \ 2{\cdot}163^c \end{array} \right\rangle$

award 1 mark per bullet

Common Error 3 : Finding angle AOT

using $\vec{OA} = \begin{pmatrix} 23 \\ 0 \\ 8 \end{pmatrix}$ *and* $\vec{OT} = \begin{pmatrix} 28 \\ -15 \\ 7 \end{pmatrix}$

- $|\vec{OA}| = \sqrt{593}$ *and* $|\vec{OT}| = \sqrt{1058}$
- $\vec{OA}.\vec{OT} = 700$
- $\left\langle \begin{array}{l} \cos(A\hat{O}T) = \dfrac{\vec{OA}.\vec{OT}}{|\vec{OA}||\vec{OT}|} \\ A\hat{O}T = 27{\cdot}9^\circ \ \ OR \ \ 0{\cdot}487^c \end{array} \right\rangle$

award 1 mark per bullet

Common Error 4 : Finding angle ABT

using $\vec{BA} = \begin{pmatrix} 35 \\ 0 \\ -1 \end{pmatrix}$ *and* $\vec{BT} = \begin{pmatrix} 40 \\ -15 \\ -2 \end{pmatrix}$

- $|\vec{BA}| = \sqrt{1226}$ *and* $|\vec{BT}| = \sqrt{1829}$
- $\vec{BA}.\vec{BT} = 1402$
- $\left\langle \begin{array}{l} \cos(A\hat{B}T) = \dfrac{\vec{OA}.\vec{OT}}{|\vec{OA}||\vec{OT}|} \\ A\hat{B}T = 20{\cdot}6^\circ \ \ OR \ \ 0{\cdot}359^c \end{array} \right\rangle$

award 1 mark per bullet

Common Error 5 : Finding angle BAT

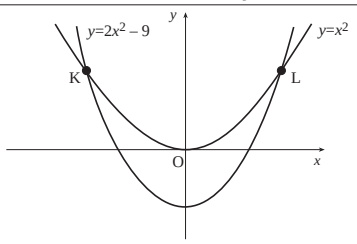
using $\vec{AB} = \begin{pmatrix} -35 \\ 0 \\ 1 \end{pmatrix}$ *and* $\vec{AT} = \begin{pmatrix} 5 \\ -15 \\ -1 \end{pmatrix}$

- $|\vec{AB}| = \sqrt{1226}$ *and* $|\vec{AT}| = \sqrt{251}$
- $\vec{AB}.\vec{AT} = -176$
- $\left\langle \begin{array}{l} \cos(B\hat{A}T) = \dfrac{\vec{AB}.\vec{AT}}{|\vec{AB}||\vec{AT}|} \\ B\hat{A}T = 108{\cdot}5^\circ \ \ OR \ \ 1{\cdot}894^c \end{array} \right\rangle$

award 1 mark per bullet

9

5 The curves with equations $y = x^2$ and $y = 2x^2 - 9$ intersect at K and L as shown.
Calculate the area enclosed between the curves. **8**



Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
5		8	C	C17	CN	05/49

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THIS GENERIC M/S MAY BE USED AS AN EQUIVALENCE GUIDE BUT ONLY WHERE A CANDIDATE DOES NOT USE THE PRIMARY METHOD OR ANY ALTERNATIVE METHOD SHOWN IN DETAIL IN THE MARKING SCHEME

- •1 ss: find intersection
- •2 pd: process quadratic equ.
- •3 ss: upper – lower
- •4 ic: interpret limits
- •5 pd: sub. & simplify Upper – Lower
- •6 pd: integrate
- •7 ic: substitute limits
- •8 pd: evaluate and complete

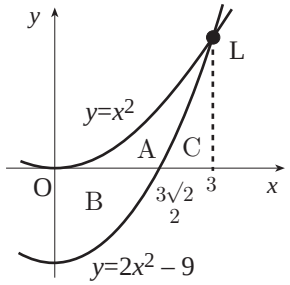
Primary Method : Give 1 mark for each •

- •1 $x^2 = 2x^2 - 9$
- •2 $x = \pm 3$
- •3 $\int upper - lower$ ***[Notes 3,4]*** **stated or implied by •5**
- •4 $eg\ \int_0^3 \ \ldots$
- •5 $x^2 - 2x^2 + 9$
- •6 $\left[-\frac{1}{3}x^3 + 9x\right]_0^3$
- •7 $\left(-\frac{1}{3}\times 3^3 + 9\times 3\right) - 0$
- •8 $2\times 18 = 36$ ***[Note 3]*** **8 marks**

Alternative 1 for •4 to •8

- •4 $eg\ \int_{-3}^3 \ \ldots$
- •5 $x^2 - 2x^2 + 9$
- •6 $\left[-\frac{1}{3}x^3 + 9x\right]_{-3}^3$
- •7 $\left(-\frac{1}{3}\times 3^3 + 9\times 3\right) - \left(-\frac{1}{3}\times(-3)^3 + 9\times(-3)\right)$
- •8 36

Alternative 2 for •3 to •8



- •3 $x = \frac{3}{2}\sqrt{2}$
- •4 $\int_0^{\frac{3}{2}\sqrt{2}} \left(9 - 2x^2\right)dx$ leading to B=$9\sqrt{2}$ (12.7)
- •5 $\int_0^3 \left(x^2\right)dx$ leading to A+C=9
- •6 $\int_{\frac{3}{2}\sqrt{2}}^3 \left(2x^2 - 9\right)dx$ leading to C=$9\sqrt{2} - 9$ (3.7) ***[Note 8]***
- •7 $A = 18 - 9\sqrt{2}$ (5.3)
- •8 $Total\ area = 36$

Notes

1 There is no penalty for working with
$\frac{1}{3}x^3 - \frac{2}{3}x^3 + 9x$ *or even* $\frac{1}{3}x^3 - \left(\frac{2}{3}x^3 - 9x\right)$ but in the latter case, the minus signs need to be dealt with correctly at some point for •5 o be awarded.

2 Candidates who attempt to find a solution using a graphics calculator earn no marks. The only acceptable solution is via calculus.

3 •3 is lost for subtracting the wrong way round and subsequently •8 may be lost for such statements as

-36
$-36\ square\ units$
$-36 = 36$
$-36\ so\ ignore\ the\ -ve$
$-36 = 36\ square\ units$

•8 may be gained for statements such as
$-36\ so\ the\ area = 36$

4 $\int_3^{-3}\left(lower - upper\right)$ *or* $\int_3^0\left(lower - upper\right)$ are technically correct and hence all 8 marks are available.

5 For $\int_K^L\left(upper - lower\right)$, •3,•5,•6 and •7 are available

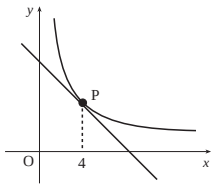
6 Differentiation loses **•6, •7 and •8**.

7 Using $x^2 + 2x^2 - 9$ and $\int_{-3}^3\left(3x^2 - 9\right)dx$ leading to zero can only gain **•4** and **•6** from the last 6 marks.

8 Candidates may attempt to split the area up.
In Alt.2, for candidates who treat "C" as a triangle , the last three marks are not available.

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6 The diagram shows the graph of $y = \frac{24}{\sqrt{x}}$, $x > 0$.



Find the equation of the tangent at P, where $x = 4$. **6**

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
6		6	B	C5, C3	CN	05/43

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- •1 ss : know to differentiate
- •2 ic : express in st. form
- •3 pd : differentiate –ve fractional index
- •4 pd : evaluate –ve fractional index
- •5 pd : evaluate y-coord
- •6 ic : state equ of tangent

Primary Method : Give 1 mark for each •

- •1 $\frac{dy}{dx} = \ldots$
- •2 $y = 24x^{-\frac{1}{2}}$
- •3 $\frac{dy}{dx} = -12x^{-\frac{3}{2}}$
- •4 $\frac{dy}{dx}_{x=4} = -\frac{3}{2}$
- •5 $y_{x=4} = 12$
- •6 $y - 12 = -\frac{3}{2}(x-4)$ ***[Notes 1,2,3]*** **6 marks**

nr $\left[2y + 3x = 36\right]$

nr = not required

Notes

1 **•4 and •6** are only available if an attempt to find the gradient is based on differential calculus.
2 **•6** is not available to candidates who find and use a perpendicular gradient.
3 **•6** is only available for a numerical value of *m*.

Common Error 1

- •1 $\frac{dy}{dx} = \ldots$
- •2 $y = 24x^{-\frac{1}{2}}$
- •3 $\frac{dy}{dx} = \frac{24x^{\frac{1}{2}}}{\frac{1}{2}}$
- •4 $\frac{dy}{dx}_{x=4} = 96$
- •5 $y_{x=4} = 12$
- •6 $y - 12 = 96(x-4)$

- •1 √
- •2 √
- •3 ×
- •4 × *eased*
- •5 √
- •6 ×√

award 4 marks

Common Error 2

- •1
- •2 $y = 24x^{-\frac{1}{2}}$
- •3 $\int 24x^{-\frac{1}{2}}dx = \frac{24x^{\frac{1}{2}}}{\frac{1}{2}} + c$
- •4 *gradient* $= 96$
- •5 $y_{x=4} = 12$
- •6 $y - 12 = 96(x-4)$

- •1 ×
- •2 √
- •3 ×
- •4 × *Note 1*
- •5 √
- •6 × *Note 1*

award 2 marks

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7 Solve the equation $\log_4(5-x)-\log_4(3-x)=2,\ \ x<3$. **4**

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
7		4	A	A7	CN	0525

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- $\bullet^1$ ss: use the log laws
- $\bullet^2$ ss: know to convert from log to expo
- $\bullet^3$ pd: process conversion
- $\bullet^4$ pd: find valid solution

Primary Method : Give 1 mark for each •

- $\bullet^1$ $\log_4\left(\frac{5-x}{3-x}\right)$
- $\bullet^2$ *use* $\log_a(b)=c \Leftrightarrow b=a^c$ **stated or implied by •3**
- $\bullet^3$ $\frac{5-x}{3-x}=4^2$ **See Cave**
- $\bullet^4$ $x=\frac{43}{15}$ **4 marks**

Notes

1 For **•4**
Accept answer as a decimal.

Common Error No.1

- $\bullet 1$ $\surd$ $\log_4\left(\frac{5-x}{3-x}\right)=\log_4(8)$
- $\bullet 2$ $\times$
- $\bullet 3$ $\times$ $\frac{5-x}{\mathbf{3}-x}=8$
- $\bullet 4$ $\times\surd$ $x=\frac{19}{7}$

award 2 marks

Common Error No.2

- $\bullet 1$ $\surd$ $\log_4\left(\frac{5-x}{3-x}\right)=2$
- $\bullet 2$ $\times$ $4^{\frac{5-x}{3-x}}=2$
- $\bullet 3$ $\times$ $\frac{5-x}{3-x}=\frac{1}{2}$
- $\bullet 4$ $\times\surd$ $x=7$ *which is not a valid sol.*

award 2 marks

Common Error No.3

- $\bullet 1$ $\surd$ $\log_4\left(\frac{5-x}{3-x}\right)=2$
- $\bullet 2$ $\times$ $\log_4\left(\frac{5-x}{3-x}\right)=\log_4 2$
- $\bullet 3$ $\times$ $\frac{5-x}{3-x}=2$
- $\bullet 4$ $\times\surd$ $x=1$.

award 2 marks

Alternative 1

- $\bullet^1$ $\log_4\left(\frac{5-x}{3-x}\right)$
- $\bullet^2$ $2\log_4 4$ **stated or implied by •3**
- $\bullet^3$ $\left(\frac{5-x}{3-x}\right)=4^2$
- $\bullet^4$ $x=\frac{43}{15}$

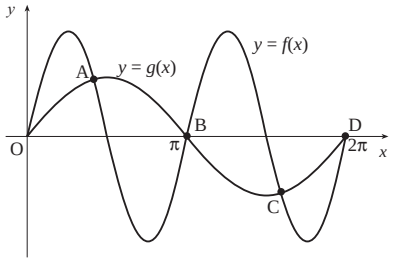
Cave

$\log_4\left(\frac{5-x}{3-x}\right)$ $\frac{5-x}{3-x}=16$ *leading to* $x=\frac{43}{15}$ *award 4 marks*	**BUT**	$\log_4\left(\frac{5-x}{3-x}\right)$ $\frac{5-x}{3-x}=2^4$ *leading to* $x=\frac{43}{15}$ $\bullet^1\surd$, $\bullet^2\times$, $\bullet^3\times$, $\bullet^4\times\surd$ *award 2 marks*

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8 Two functions, f and g, are defined by $f(x) = k\sin(2x)$ and $g(x) = \sin(x)$ where $k > 1$.

The diagram shows the graphs of $y = f(x)$ and $y = g(x)$ intersecting at O, A, B, C and D. Show that, at A and C, $\cos(x) = \frac{1}{2k}$.



5

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
8		5	A	T10	CN	05/47

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- •1 ss: equate for intersection
- •2 ss: use double angle formula
- •3 pd: factorise
- •4 pd: process two solutions
- •5 ic: complete proof

Primary Method : Give 1 mark for each •

- •1 $k\sin(2x) = \sin(x)$ ***[Note 1]***
- •2 $k \times 2\sin(x)\cos(x)$
- •3 $\sin(x)\big(2k\cos(x) - 1\big) = 0$
- •4 $\sin(x) = 0$ *and* $\cos(x) = \frac{1}{2k}$
- •5 $\sin(x) = 0 \Rightarrow x = 0, \pi, 2\pi$ i.e. at (O),B and D ***[Note 2]*** *and* $\cos(x) = \frac{1}{2k}$ is for A and C.

5 marks

Notes

1 Only **•1** is available for candidates who substitute a numerical value for k at the start.

2 **•5** is only available if a suitable comment regarding points (O), B and D is made.

3 If all the terms are transposed to one side, then an "=0" needs to appear at least once.

4 For Alternative 3
•4 and **•5** are not available unless **•3** has been awarded.

Common Error 1

- •1 √ $k\sin(2x) = \sin(x)$
- •2 √ $k \times 2\sin(x)\cos(x) - \sin(x) = 0$
- •3 √ $\sin(x)\big(2k\cos(x) - 1\big)$
- •4 × $2k\cos(x) - 1 = 0$
- •5 × $\cos(x) = \frac{1}{2k}$ at A and C.

award 3 marks

Common Error 2

- •1 √ $k\sin(2x) = \sin(x)$
- •2 √ $k \times 2\sin(x)\cos(x) = \sin(x)$
- •3 × $k \times 2\cos(x) = 1$
- •4 ×
- •5 × $\cos(x) = \frac{1}{2k}$ at A and C.

award 2 marks

Alternative 1 for •4 and •5

- •4 at (O), B and D, $\sin(x) = 0$
- •5 so at A and C, $2k\cos(x) - 1 = 0$ $\Rightarrow \cos(x) = \frac{1}{2k}$.

Alternative 2 for •4 and •5

- •4 at A and C, $\sin(x) \neq 0$
- •5 so at A and C, $2k\cos(x) - 1 = 0$ $\Rightarrow \cos(x) = \frac{1}{2k}$.

Alternative 3 for •1 to •5

- •1 $k\sin(2x) = \sin(x)$
- •2 $k \times 2\sin(x)\cos(x) = \sin(x)$
- •3 at A and C, $\sin(x) \neq 0$
- •4 so at A and C, $2k\cos(x) = 1$
- •5 $\cos(x) = \frac{1}{2k}$

13

9 The value V (in £ million) of a cruise ship t years after launch is given by the formula $V = 252e^{-0{\cdot}06335t}$.

(a) What was its value when launched? **1**

(b) The owners decide to sell the ship once its value falls below £20 million. After how many years will it be sold? **4**

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
9	a	1	B	A34	CN	05/76
	b	4	A	A34	Ca	

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- •1 pd: evaluate at $t = 0$
- •2 ic: substitute $V = 20$
- •3 pd: process
- •4 ic: expo to log conversion
- •5 pd: solve a logarithmic equation

Primary Method : Give 1 mark for each •

- •1 $V_{t=0} = 252\,(£m)$ **1 mark**
- •2 $252e^{-0{\cdot}06335t} = 20$
- •3 $e^{-0{\cdot}06335t} = \frac{20}{252}$
- •4 $-0{\cdot}06335t = \ln\left(\frac{20}{252}\right)$
- •5 $t = 40$ ***[Note 1]*** **4 marks**

Notes

in (b)

1 For **•5** accept any correct answer which rounds to 40.
2 An answer obtained by trial and improvement which rounds to 40 may be awarded a max. of 1 mark (out of 4) **but only** if they have checked 39 as well.
3 In following through from an error, •5 is only available for a positive answer.

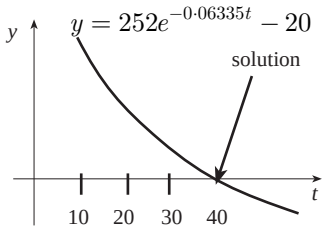
Common Error 1

- •2 √ $\log\left(252e^{-0{\cdot}06335t}\right) = \log 20$
- •3 × $-0.06335t\log 252 = \log 20$
- •4 × $-0.06335t = \dfrac{\log 20}{\log 252}$
- •5 × $t = -8.55$

award 1 mark

Solution via graphics calculator

- •2 $252e^{-0{\cdot}06335t} = 20$
- •3 *choose to graph* $y = 252e^{-0{\cdot}06335t} - 20$
- •4 *a sketch [see below]*
- •5 $t = 40$



Note

You could also graph, for example, $y = 252e^{-0{\cdot}06335t}$ *and* $y = 20$

Alternative 1 for (b) (takings logs of both sides)

- •2 $252e^{-0{\cdot}06335t} = 20$
- •3 $e^{-0{\cdot}06335t} = \frac{20}{252}$
- •4 $-0{\cdot}06335t\log_k(e) = \log_k\left(\frac{20}{252}\right)$ *where* $k = e$ *or* $k = 10$
- •5 $t = 40$ ***[Note 1]***

Alternative 2

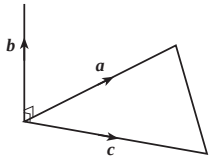
- •2 $252e^{-0{\cdot}06335t} = 20$
- •3 $\log 252 - 0.06335t\log e = \log 20$
- •4 $5.53 - 0.06335t = 2.99$
- •5 $t = 40$

Alternative 3

- •2 $252e^{-0{\cdot}06335t} = 20$
- •3 $\ln 252 + \ln e^{-0.06335t} = \ln 20$
- •4 $-0.06335t\ln e = \ln 20 - \ln 252$
- •5 $t = 40$

14

10 Vectors a and c are represented by two sides of an equilateral triangle with sides of length 3 units, as shown in the diagram. Vector b is 2 units long and b is perpendicular to both a and c. Evaluate the scalar product $\boldsymbol{a}.(\boldsymbol{a}+\boldsymbol{b}+\boldsymbol{c})$.



4

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
10		4	A	G29	CN	05/31

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- •1 ss: use distributive law
- •2 pd: process scalar product
- •3 pd: process scalar product
- •4 pd: process scalar product & complete

Primary Method : Give 1 mark for each •

- •1 $\boldsymbol{a.a}+\boldsymbol{a.b}+\boldsymbol{a.c}$ **see CAVE** *[Notes 1,2]*
- •2 $\boldsymbol{a.a} = 9$
- •3 $\boldsymbol{a.c} = \frac{9}{2}$ *[Note 3]*
- •4 $\boldsymbol{a.b} = 0$ *and a total of* $13\frac{1}{2}$

4 marks

Notes

1 Treat $\underline{a}.\underline{a}$ written as a^2 as bad form.

2 Treat $\underline{a}.\underline{b}$ written as ab as an error unless it is subsequently evaluated as a scalar product. Similarly for $\underline{a}.\underline{c}$.

3 Using $\underline{p}.\underline{q} = \mid p \mid\mid q \mid \sin\theta$ consistently loses 1 mark. (ie max. available is 3)

4 When attaching the components

$\boldsymbol{c} = \begin{pmatrix}3\\0\\0\end{pmatrix}, \boldsymbol{b} = \begin{pmatrix}0\\0\\2\end{pmatrix}, \boldsymbol{a} = \begin{pmatrix}\frac{3}{2}\\\frac{3\sqrt{3}}{2}\\0\end{pmatrix}$, all marks are available.

When attaching the components

$\boldsymbol{c} = \begin{pmatrix}3\\0\\0\end{pmatrix}, \boldsymbol{b} = \begin{pmatrix}0\\0\\2\end{pmatrix}, \boldsymbol{a} = \begin{pmatrix}0\\3\\0\end{pmatrix}$,only **•1** is available.

CAVE

$\boldsymbol{a.(a+b+c)} = \boldsymbol{a.a}+\boldsymbol{a.b}+\boldsymbol{a.c}$
followed by
$\boldsymbol{a.a} = 9$
earns •1 *and* •2.

but

$\boldsymbol{a.(a+b+c)} = \boldsymbol{a.a}+\boldsymbol{a.b}+\boldsymbol{a.c}$
followed by
$\boldsymbol{a.a} = 9,\ \boldsymbol{a.c} = 9,\ \boldsymbol{a.b} = 6$
earns •1 *only.*

11 (a) Show that $x = -1$ is a solution of the cubic equation $x^3 + px^2 + px + 1 = 0$. **1**

(b) Hence find the range of values of p for which all the roots of the cubic equation are real. **7**

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
11	a	1	C	A21	CN	05/54
	b	7	A	A22	CN	

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- $\bullet^1$ pd: evaluate the function at $x = -1$
- $\bullet^2$ ss: strategy for finding other factors
- $\bullet^3$ ic: quadratic factor
- $\bullet^4$ ss: strategy for real roots
- $\bullet^5$ ic: substitute
- $\bullet^6$ pd: process
- $\bullet^7$ ss: starts to solve inequation
- $\bullet^8$ ic: complete

Primary Method : Give 1 mark for each •

$\bullet^1$ $f(-1) = -1 + p - p + 1 = 0$ **1 mark**

$\bullet^2$

$$-1 \left| \begin{array}{cccc} 1 & p & p & 1 \\ & -1 & 1-p & -1 \end{array} \right.$$
$$1 \quad p-1 \quad 1 \quad 0$$

$\bullet^3$ $x^2 + (p-1)x + 1 = 0$ ***[Note 2]***

$\bullet^4$ "$b^2 - 4ac$" *and* "≥ 0" ***[Notes 3,4]***

$\bullet^5$ $(p-1)^2 - 4$

$\bullet^6$ $(p-3)(p+1)$

$\bullet^7$ $p = 3, p = -1$

$\bullet^8$ $p \leq -1, p \geq 3$ ***[Note 6]*** **7 marks**

Notes

1 For alternative method 1, **•2**
•2 (as is •3 also) is for interpreting the result of a synthetic division.
Candidates must show some acknowledgement of the result of the synthetic division. Although a statement w.r.t. the zero is preferable, accept something as simple as "underlining" the zero.

2 Treat "= 0" missing at **•3** as Bad Form

3 •4 is only available as a consequence of obtaining a quadratic factor from a division of the cubic.

4 Using $b^2 - 4ac > 0$ loses **•4**
An "≥" must appear at least once somewhere between •4 and •6

5 Where errors occur at the •3/•5 stages, then •6,•7,•8 are still available for solving a '3-term' quadratic inequation.

6 Evidence for **•8** may be a table of values or a sketch

7 For candidates who start with $\dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, all marks are available (subject to working being equivalent to the Primary Method).

8 Wrong disciminant:
Using $b^2 + 4ac$ only **•5** (out of the last 5 marks) is available.
Any other expression masquerading as the discriminant loses all of the last 5 marks.

Alternative method 1 for marks 1,2 (starting with synth. division)

$\bullet^1$

$$-1 \left| \begin{array}{cccc} 1 & p & p & 1 \\ & -1 & 1-p & -1 \end{array} \right.$$
$$1 \quad p-1 \quad 1 \quad 0$$

$\bullet^2$ $f(-1)=0$ ***[Note 1]***

etc

Marks should still be recorded as out of 1 and 7

Alternative method 2 for marks 1,2 (quad. factor obtained by inspection)

$\bullet^1$ $f(-1) = -1 + p - p + 1 = 0$

$\bullet^2$ $f(x) = (x+1)(x^2 \ \ldots\ldots)$

etc

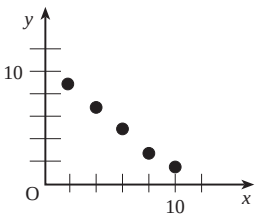
Common Error 1 (marks 5 to 8)

$(p-1)^2 - 4 \geq 0$
$(p-1)^2 \geq 4$
$p - 1 \geq 2$
$p \geq 3$

award 2 marks out of last 4

16

[4] The scatter diagram shows 5 pairs of data values for x and y where $\Sigma x = 30,\ \Sigma y = 26,\ \Sigma x^2 = 220,\ \Sigma y^2 = 168$ and $\Sigma xy = 120$.



(a) Find the equation of the regression line. **4**

(b) Estimate the value of y when $x = 5$. **1**

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
S1		4	C	4.4.2	Ca	05/76
	b	1	C	4.4.2	CN	

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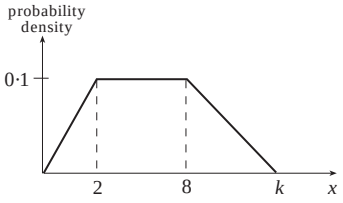
- $\bullet^1$ pd: calculate S_{xy}
- $\bullet^2$ pd: calculate S_{xx}
- $\bullet^3$ pd: calculate b
- $\bullet^4$ pd: calculate a & state equ.
- $\bullet^5$ ic: use equ. of regression line

Primary Method : Give 1 mark for each •

- $\bullet^1$ $S_{xy} = -36$
- $\bullet^2$ $S_{xx} = 40$
- $\bullet^3$ $b = -0\cdot 9$
- $\bullet^4$ $a = 10\cdot 6$ *and* $y = 10\cdot 6 - 0\cdot 9x$ **4 marks**

- $\bullet^5$ $y_{x=5} = 6.1$ **1 mark**

[7] The diagram represents the probability density function for a continuous random variable X.



(a) Find the value of k. 3

(b) Find the median. 2

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
S4	a	3	A	4.3.1	CN	05/83
	b	2	A	4.3.5	CN	

The primary method m/s is based on the following generic m/s. THIS GENERIC M/S MAY BE USED AS AN EQUIVALENCE GUIDE BUT ONLY WHERE A CANDIDATE DOES NOT USE THE PRIMARY MWETHOD OR ANY ALTERNATIVE METHOD SHOWN IN DETAIL IN THE MARKING SCHEME

- $\bullet^1$ ss: state total area = 1
- $\bullet^2$ ic: find expression for total area
- $\bullet^3$ pd: process

- $\bullet^4$ ss: know total area = 0.5
- $\bullet^5$ pd: process

Primary Method : Give 1 mark for each •

- $\bullet^1$ $area = 1$
- $\bullet^2$ $0\cdot 1 + 0\cdot 6 + \frac{1}{2}\left(k-8\right)\times 0.1$
- $\bullet^3$ $k = 14$ **3 marks**

- $\bullet^4$ $0\cdot 1 + \left(m-2\right)\times 0.1 = \frac{1}{2}$
- $\bullet^5$ $m = 6$ **2 marks**

[9] (*a*) Explain briefly the difference between sample standard deviation and range as measures of spread. 1

(*b*) In statistics mode, a calculator shows the summary statistics for a certain data set.
One data value, 1·2, is shown to be erroneous and is deleted.
Calculate the sample standard deviation of the new data set of 19 values correct to 3 decimal places. 4

$\bar{x} =$	2.325
$S_x =$	0.573883355
$\sigma_x =$	0.559352304
$\Sigma x =$	46.5
$\Sigma x^2 =$	114.37
$n =$	20
$x_{min} =$	1.2
$x_{max} =$	3.2

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
S3	a	1	B	4.2.11/12	CN	05/79
	b	4	B	4.1.1	Ca	

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- $\bullet^1$ ic: explanation
- $\bullet^2$ pd: find new $\sum x$
- $\bullet^3$ pd: find new $\sum x^2$
- $\bullet^4$ ss: use formula for S_x
- $\bullet^5$ pd: process

Primary Method : Give 1 mark for each •

- $\bullet^1$ SD is a measure of spread about mean whereas $(x_{max} - x_{min})$ is a measure of range. **1 mark**
- $\bullet^2$ $\Sigma x = 45.3$
- $\bullet^3$ $\Sigma x^2 = 112.93$
- $\bullet^4$ $S = \sqrt{\frac{1}{18}\left(112.93 - \frac{45.3^2}{19}\right)}$
- $\bullet^5$ $0 \cdot 523$ **4 marks**

[10] A large organisation decides to run a mini-lottery for charity.

- Each participant selects any three different numbers from 1 to 20 inclusive.
- Every Friday the three winning numbers are drawn at random from the 20.
- Each participant with these winning numbers shares the jackpot.

(*a*) Find the number of possible combinations and hence find the probability of a particular combination winning a share of the jackpot. **2**

(*b*) Find the probability that someone chooses the winning combination exactly twice within 3 successive weeks. **3**

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
S4	a	2	B	4.2.5, 4.2.3	Ca	05/78
	b	3	A	4.2.7	Ca	

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- $\bullet^1$ ss: find combination
- $\bullet^2$ pd: calculate probability
- $\bullet^3$ ic: interpret p(win)
- $\bullet^4$ ss: find combination
- $\bullet^5$ pd: process

Primary Method : Give 1 mark for each •

- $\bullet^1$ No. of outcomes $= \binom{20}{3}$
- $\bullet^2$ $prob = \frac{1}{\binom{20}{3}} = \frac{1}{1140}$ **2 marks**
- $\bullet^3$ $p(L) = \frac{1139}{1140}$
- $\bullet^4$ $p(2 \text{ wins in } 3)$
  $= 3 \times \left(\frac{1}{1140}\right)^2 \times \left(\frac{1139}{1140}\right)$
- $\bullet^5$ 2.306×10^{-6} **3 marks**

18